

Use these Open Source Libraries for Graphical Representation and Visualisation



The hunt for data visualisation tools that can present huge data sets on Web based platforms can be tedious, since a number of software frameworks and tools are available for plotting data and for dynamic graph generation. In this article, we will cover the free and open source resources that can be used for the visualisation of big data sets.

The libraries and toolkits discussed in this article can be used for rendering dynamic plots on desktop, mobile and Web based platforms so that a quick summary of results can be presented. These tools can be used by data scientists and researchers for an effective analysis of dynamic data.

Key features of plotting libraries

The key features and characteristics that are generally required for the tasks associated with data science, analytics and knowledge discovery, and are directly related to effective visualisation and plotting are:

- Free and open source without any licensing issues
- Programmer-friendly drag and drop editor
- Support for assorted Web standards
- Animated charts and plots for better analysis of data
- Integrated wizards and templates
- Data imports from multiple sources

- Integration of application programmer interfaces (APIs) to third party channels
- Responsive outputs
- Multi-colour plots with multiple dimensional views

Use cases and features for Web applications

The real-world scenarios of desktop and Web based applications need different types of visual components so that the application can be built with a user-friendly interface. To develop such applications with high performance and better understanding, there is a need to integrate the visualisation and plotting modules.

A few scenarios and use cases where the need comes up for plotting, visualisation and dynamic graphs in software applications are:

- Real-time maps and street views for mobile app based delivery systems (examples are Zomato, Swiggy, Uber Eats, Amazon, etc)

- Dynamic graphs and plots for predictions (examples are the stock market, health services, e-governance, commodity prices, and weather forecasting)
 - Knowledge discovery and predictive mining (examples are machine learning algorithms and time series)
- A few popular, free, open source and online libraries for graphical representations and visualisations are listed below.

They are quite powerful and widely used for Web based applications and multi-dimensional views of data. These platforms can be used to view, analyse, and evaluate big as well as massive data sets.

Candela

URL: <https://candela.readthedocs.io/en/latest/index.html>

Candela is certainly one of the best free and open source data visualisation packages. It can be integrated into the data science tasks for visualisation using multiple programming interfaces, including JavaScript, Python and R. Candela supports assorted plots and graphical segments, including

